

## Wörndle Interdisciplinary Project – Topic

### **„Directors‘ dealings behavior of managers of top US high-tech firms trading on Nasdaq after the US presidential election 2024 and the quality of corresponding ChatGPT calculations“**

#### **Tasks**

The topic is based on chapter 8.2 in PEF\_8 (Project and Enterprise Financing) (pages 32-55). In this project directors‘ dealings (DD) in shares of 30 US high-tech companies included in the Nasdaq Composite index should be analyzed regarding frequency and performance of DD transactions. Time period: November 2024 until Mai 2025. The DD of interest are pure purchase and pure sale transactions in shares by managers of the company they are working at. The data can be found on the Web-Site:

**<http://www.openinsider.com/>**

It contains a summary of SEC Form 4 Filings of (legal) insider transactions.

Starting references/literature will be provided after agreement regarding the topic.

The 30 US high-tech companies are (in brackets (for some) the ticker symbol):  
Microsoft (MSFT), NVIDIA (NVDA), Apple (AAPL), Amazon (AMZN), Alphabet (GOOG), Meta(META), Tesla (TSLA), Broadcom (AVGO), Oracle (ORCL), Palantir (PLTR), Salesforce (CRM), Intel, IBM, Adobe, PayPal, ServiceNow, Cisco, Dell, Micron Technology, Automatic Data Processing, Texas Instruments, Applied Materials, VMware, eBay, Analog Devices, Lam Research, Verizon Communications, CrowdStrike (CRWD), Arista Networks Inc (ANET), AppLovin Corporation (APP)

In addition to generating the analysis via your own analyses and calculations, the performance of ChatGPT regarding DD transactions should be analyzed. For each manager of the above companies that undertook a DD transaction between November 2024 and Mai 2025 ChatGPT should be asked for a list of all DD transaction the corresponding person has performed in this period. Then, based on this list, a second question to ChatGPT is to calculate the performance (cumulative abnormal returns, see below) for the DD transactions.

**(Step 1)** Collection of **all** directors‘ dealings (DD) of the type „Purchase“ and „Sale“ (*Transaction Filing*) for the **period** November 2024 until Mai 2025 (i.e. *Section Date: Filing Date: November 2024 until Mai 2025*) for each of the above 30 companies. Here is an example for the company Adobe with the Ticker Symbol ADBE:

**OpenInsider** Latest Top Charts Tickers Go

**ADBE - Adobe Inc. - SEC Form 4 Insider Trading Screener**  
 Services - Business Services - Prepackaged Software - ADBE (796343)

**General**  
 Tickers: ADBE  
 Insider: Insider Name or CI#  
 Sh Price \$ Min Max  
 Lqdy M\$ Min Max

**Date**  
 Filing Date: All dates  
 Filing Date Range:  
 Trade Date: All dates  
 Trade Date Range:  
 Filing Delay: M M  
 N Days Ago:

**Transaction Filing**  
☒ P - Purchase ☐ F - Tax  
☒ S - Sale ☐ M - Option Ex  
☐ A - Grant ☐ X - Option Ex  
☐ D - Sale to Iss ☐ C - Crv Deriv  
☐ G - Gift ☐ W - Inherited  
☐ No deriv ☐ Multiple Days  
 Traded K\$ Min Max  
 Own Chg % Min Max

**Industry**  
 All Sectors (except Funds)

**Insider Title**  
☐ Officer  
☐ COB  
☐ CEO  
☐ Pres  
☐ COO  
☐ CFO  
☐ GC  
☐ VP  
☐ Director  
☐ 10%own  
☐ Other

Group by: Filing  
 Sort by: Filing Date  
 Max Results: 1000  
 Page: 1  
 Search clear

And part of the result:

701 results. Finviz - SEC - Yahoo - Stockcharts - Tradingview  
 TC cnt 1d 1w 1m 6m  
 P -5 -0 +1 -1 +22  
 S 530 +0 +0 +2 +8

| X | Filing Date         | Trade Date | Ticker | Insider Name     | Title                        | Trade Type   | Price    | Qty     | Owned   | ΔOwn | Value        |
|---|---------------------|------------|--------|------------------|------------------------------|--------------|----------|---------|---------|------|--------------|
|   | 2022-02-01 17:45:06 | 2022-01-28 | ADBE   | Rao Dana         | EVP, Gen. Counsel, Secretary | S - Sale     | \$507.25 | -5,300  | 17,776  | -23% | -\$2,688,425 |
| D | 2022-01-26 17:12:50 | 2022-01-25 | ADBE   | Lewnes Ann       | EVP, CMO, Corp Strategy, Dev | S - Sale+OE  | \$507.63 | -10,004 | 25,672  | -28% | -\$5,078,348 |
|   | 2022-01-19 17:47:39 | 2022-01-19 | ADBE   | Warnock John E   | Dir                          | S - Sale     | \$518.73 | -4,000  | 429,662 | -1%  | -\$2,074,938 |
| M | 2022-01-10 17:33:15 | 2022-01-06 | ADBE   | Desmond Laura    | Dir                          | P - Purchase | \$513.64 | +973    | 28,979  | +3%  | +\$499,770   |
|   | 2021-12-23 16:46:11 | 2021-12-23 | ADBE   | Chen Gloria      | EVP, Chief People Officer    | S - Sale     | \$568.96 | -42     | 12,846  | 0%   | -\$23,896    |
|   | 2021-12-21 18:28:17 | 2021-12-20 | ADBE   | Garfield Mark S. | SVP, CAO, Corp. Controller   | S - Sale     | \$545.61 | -474    | 1,108   | -30% | -\$258,619   |
|   | 2021-12-21 18:28:07 | 2021-12-20 | ADBE   | Belsky Scott     | EVP, CPO, Creative Cloud     | S - Sale     | \$545.61 | -2,750  | 21,497  | -11% | -\$1,500,428 |
|   | 2021-11-24 19:08:02 | 2021-11-24 | ADBE   | Chen Gloria      | EVP, Chief People Officer    | S - Sale     | \$654.73 | -42     | 12,775  | 0%   | -\$27,499    |
| D | 2021-10-26 17:12:09 | 2021-10-26 | ADBE   | Rao Dana         | EVP, Gen. Counsel, Secretary | S - Sale+OE  | \$650.91 | -489    | 14,579  | -3%  | -\$318,295   |

Attached is also the full result of this example downloaded in an MS-Excel file (*DD\_Example - Adobe.xlsx*).

As the max number of downloads is limited by 1000 it is best to download the DD information company by company (if necessary; students also often use a web-scraper for this task). The key is the ticker symbol (e.g. ADBE for Adobe). An Excel file of companies included in the Nasdaq Composite index will be provided (after agreement regarding the topic).

The Ticker Symbol is the key to find DD for the corresponding company and is to be inserted in the field *General: Tickers*.

**(Step 2)** Calculate daily Abnormal Returns (ARs) for the period  $\pm 20$  trading days around each DD transaction (event day  $t=0$  is the announcement day of the transaction), based on the Market Model Adjusted Return method (see e.g. PEF\_8 pages 32-40 as well as the start literature).

- ➔ If several DD of the same person take place on the same day, then these transactions should be condensed to one transaction.
- ➔ As soon as steps 1 and 2 are finished Datastream total return indices will be provided for each company. These total return indices can then directly be used to calculate daily

returns. They include normal as well as all artificial share price changes (i.e., e.g., dividend payments, correction factors due to stock splits or SEOs, etc.).

**(Step 3)** Calculate Cumulative Abnormal Returns (CARs) for the periods -20 to -1 trading days, 0 to 1 trading days, 0 bis 5 trading days, and 0 to 10 trading days, using the ARs calculated in step 3.

➔ Include these CARs in the Table generated in step 1.

**(Step 4)** Use ChatGPT to generate a list of all DD transactions of those managers that undertook a DD transaction based on the Web-Database above (period: November 2024 – Mai 2025). Analyze whether ChaptGPT is able to find all DD transactions for the above 30 companies.

**(Step 5)** Use ChatGPT to calculate the cumulative abnormal returns (similar to Step 3 (i.e. for the periods -20 to -1 trading days, 0 to 1 trading days, 0 bis 5 trading days, 0 to 10 trading days, and 0 to 20 trading days)) for the DD transactions ChatGPT presents in Step 4.

**(Step 6)** Cross sectional analyses: Analyze the distribution of CARs (i.e., all 30 companies, all transactions) for the groups:

- (5a) All purchases (your calculations)
- (5b) All sales (your calculations)
- (5c) All purchases (calculations of ChatGTP)
- (5d) All sales (calculations of ChatGTP)

➔ Calculation of important distribution parameters of CARs for each of these groups: mean, median, standard deviation, minimum, maximum

➔ Significance test for each group for the following null hypotheses:  
(i) CAR-mean = 0, (ii) CAR-median = 0

Use the following significance test:

\* Non-parametric test: Generalized Rank Test based on Kolari and Pynnönen (2011), pp. 955-956 especially.

➔ Comparison of groups 5a and 5c, as well as 5b and 5d based on mean and median values. Are the differences in (i) means of CARs and (ii) medians of CARs of the groups (i.e., 5a and 5c, as well as 5b and 5d) significant different from each other? Perform corresponding significance tests.

➔ The analysis should also include a section with descriptive statistics of the data sample used (i.e. the DD and the returns), including the process of generating the final data sample (based on data availability).